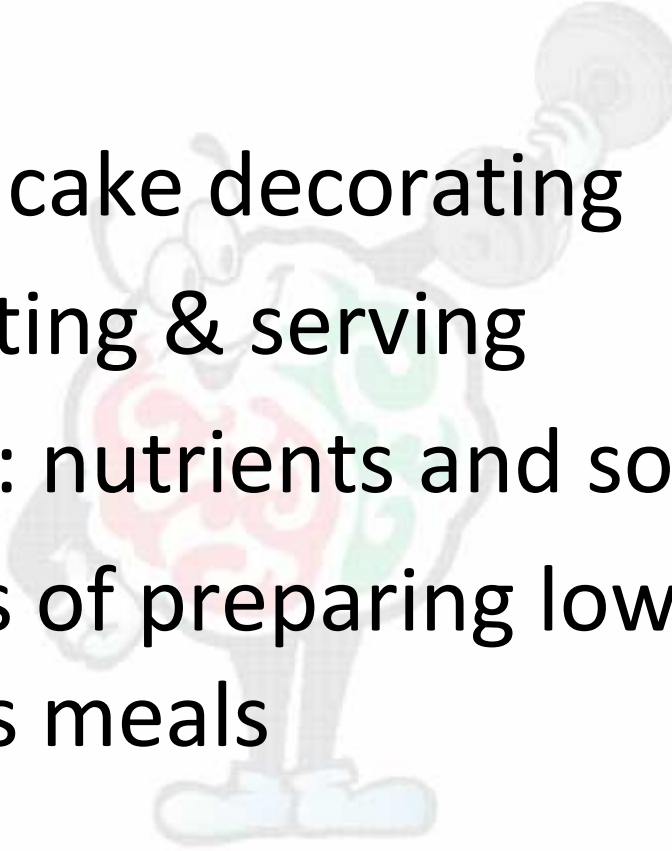


Technology and Livelihood Education: FOODS

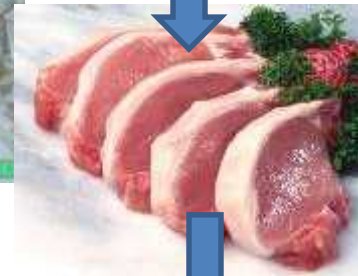
Rizza Marie T. Magbuhat
Faculty, College of Home Economics
UP-Diliman

Competencies:

1. Food selection, preparation and cooking
 2. Baking & cake decorating
 3. Table setting & serving
 4. Nutrition: nutrients and sources
 5. Principles of preparing low cost & nutritious meals
- 

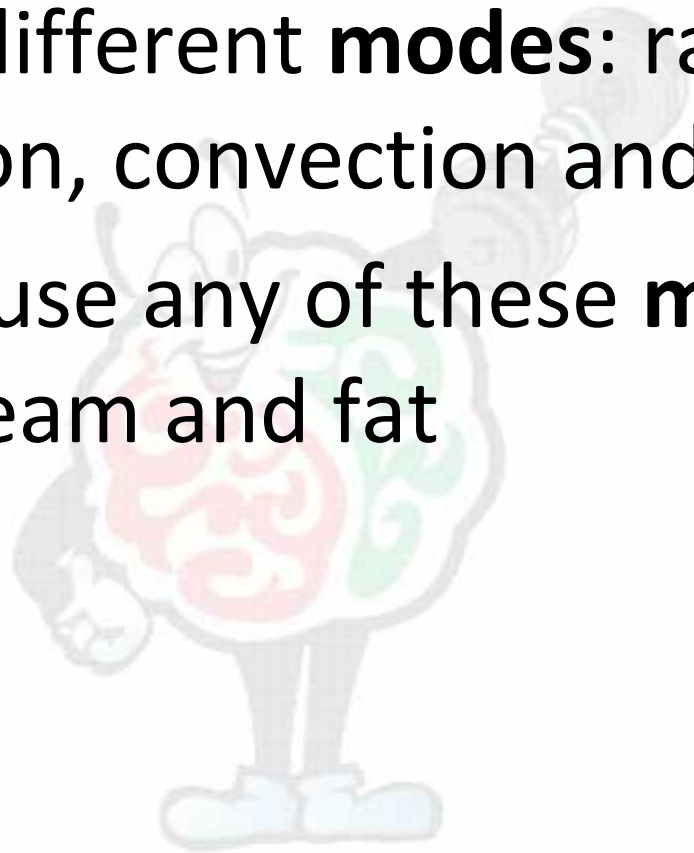
Kinds of food

- Source: plant and animal
- Level of processing:
 - primary processed
 - preserved
 - manufactured



Cooking

- Transfer of heat energy
 - through different **modes**: radiation, conduction, convection and induction;
 - and may use any of these **media**: air, water, steam and fat



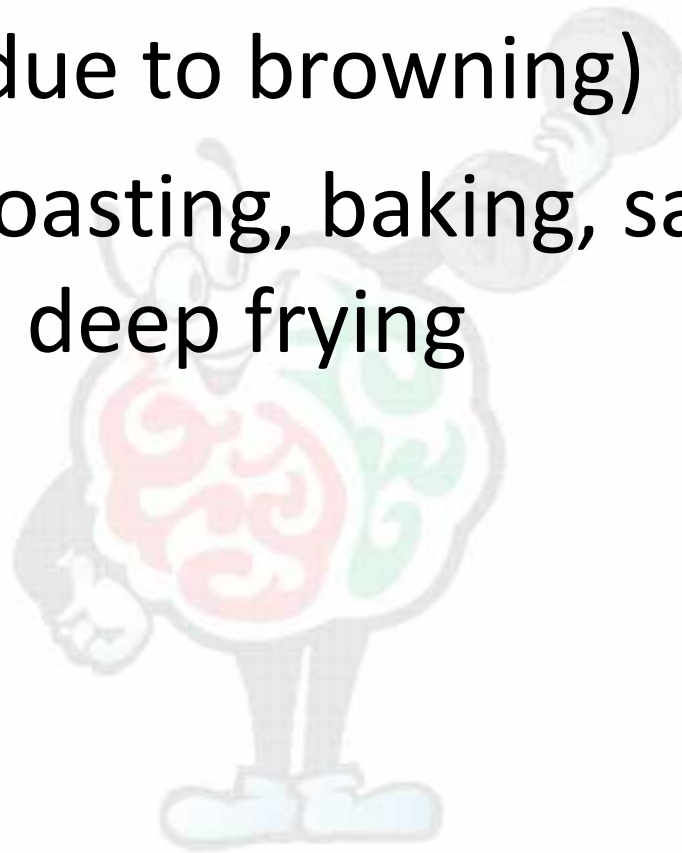
(Bennion and Scheule, 2000 and De Leon SY, 1999)

Contrast:



Cooking methods and their cooking media

- **Dry heat cooking methods:** using air or fat (rich flavor due to browning)
 - Broiling, roasting, baking, sautéing, pan frying and deep frying



Cooking methods and their cooking media

- **Moist heat cooking methods:** using water or steam (tenderize and emphasize natural flavors)
 - Poaching, simmering, boiling and steaming
- **Combination cooking methods**
 - braising and stewing

Food

- Anything that people eat to provide nourishment (nutrients)

1. Water

2. Carbohydrates

3. Protein

4. Fat

5. Vitamins

6. Minerals



WATER

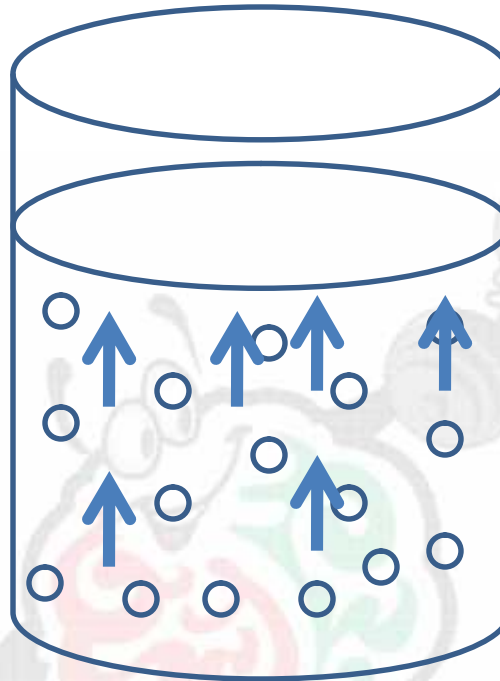
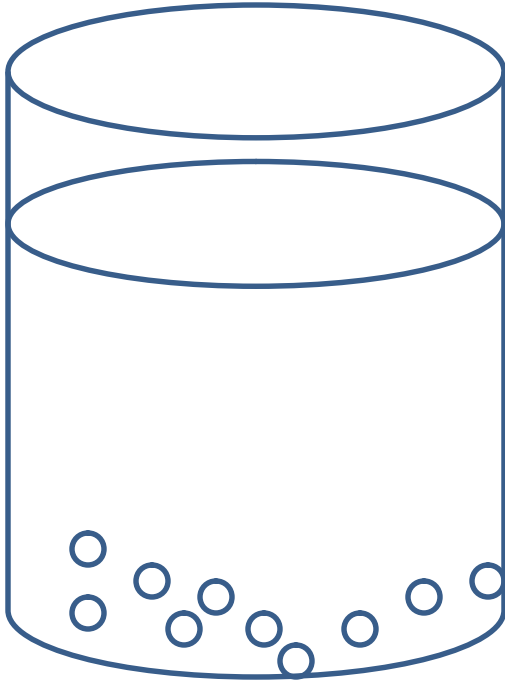
- Solvent
- Medium for heat transfer
- Found in beverages and in food



Water as a medium of heat transfer

POACH

(71 to 82 ° C)

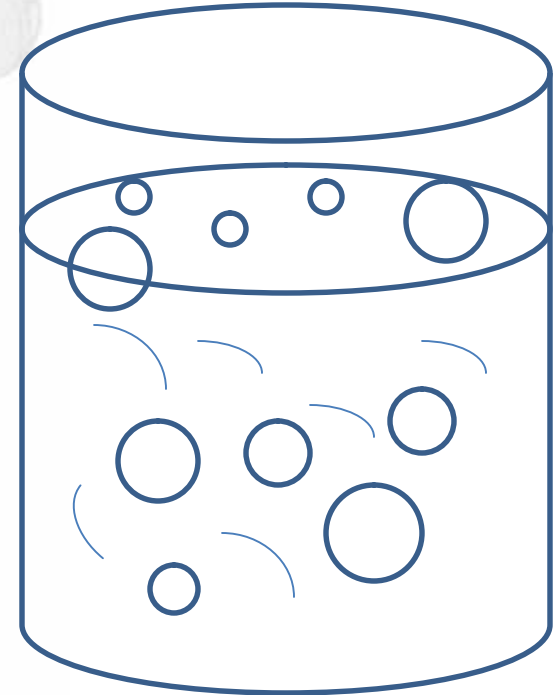


SIMMER

(82 - less than 100° C)

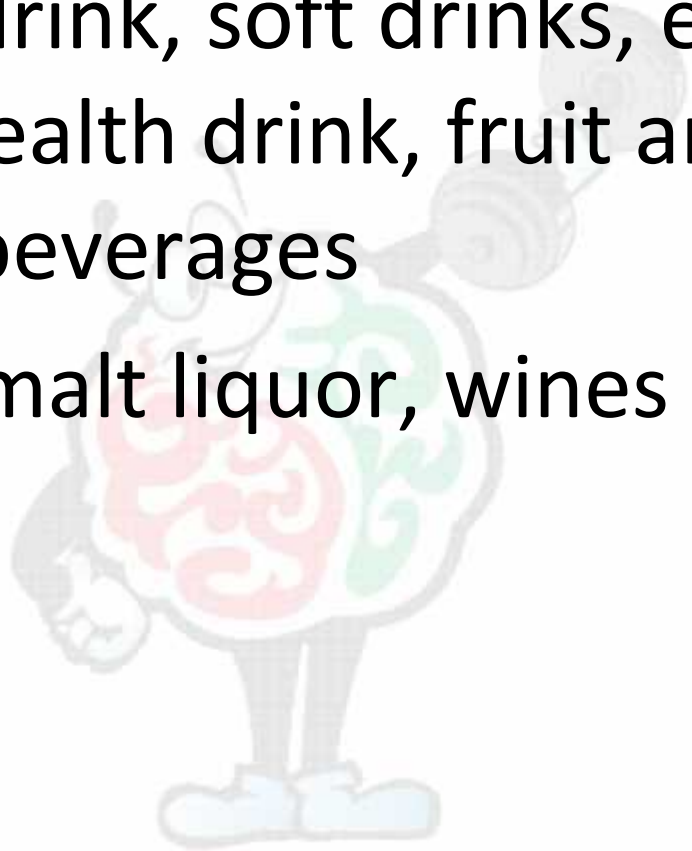
BOIL

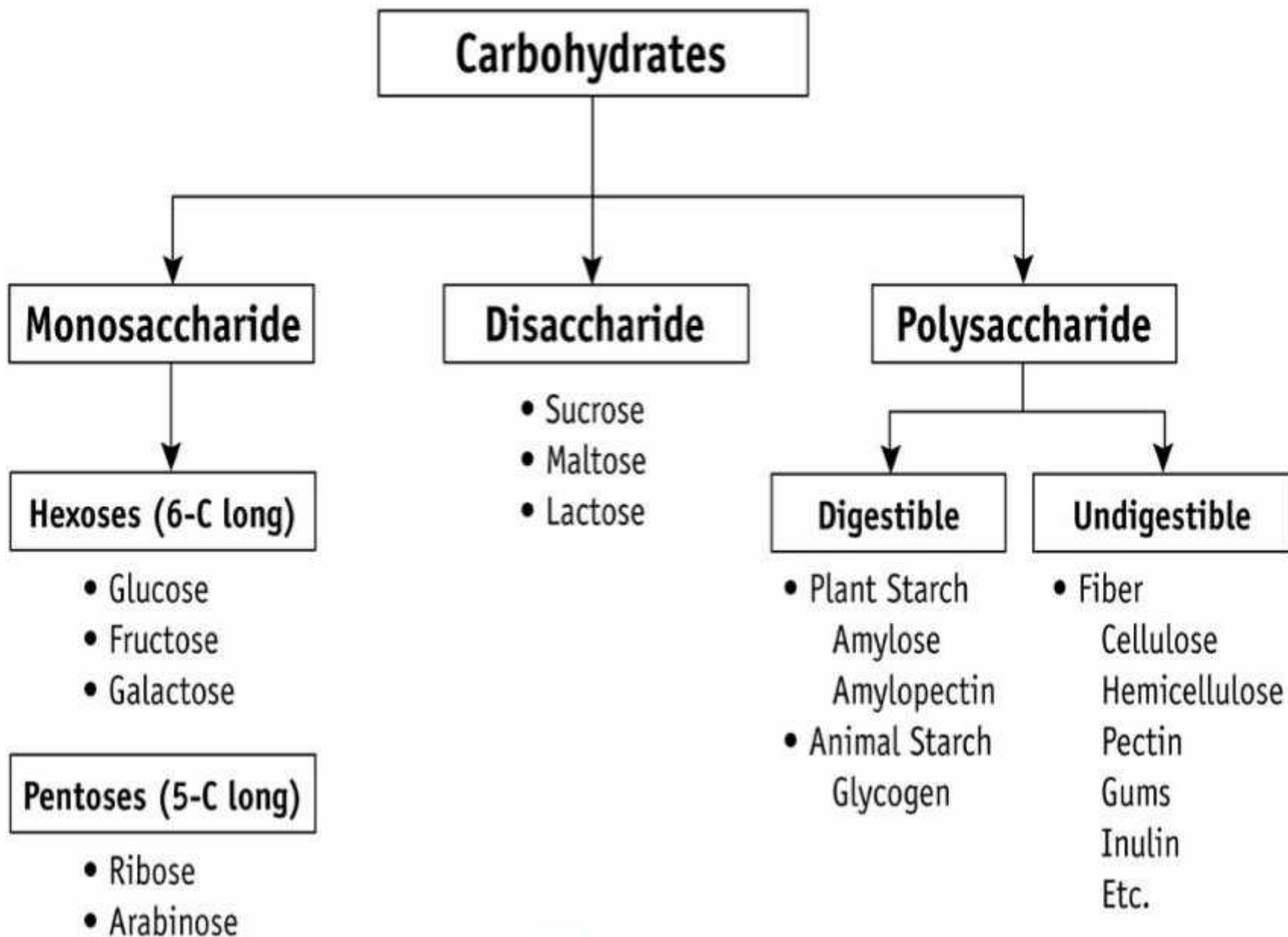
(100° C)



Water in Beverages

- **Non-alcoholic:** water, coffee, tea, chocolate drink, soft drinks, energy, sports & health drink, fruit and vegetable beverages
- **Alcoholic:** malt liquor, wines and spirits





Crystalline sugar

- Panutsa (sinakab or binagol)
- Raw brown sugar (muscovado sugar or kinugay)
- Brown sugar
- Washed sugar
- Granulated sugar (table or refined)
- Loaf sugar & lump sugar



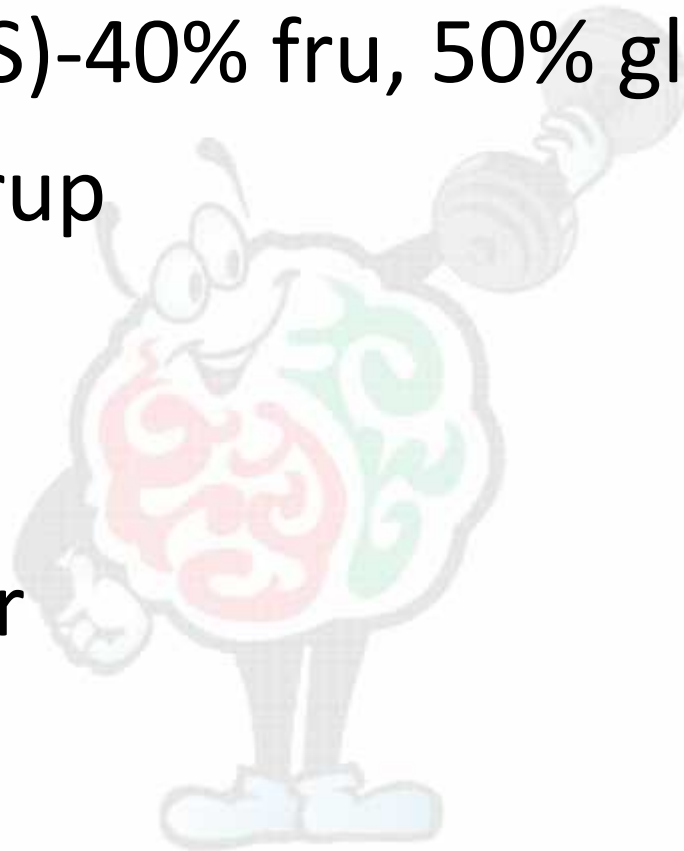
Crystalline sugar

- Sanding sugar
- Powdered sugar (confectioner's, caster and icing)



Liquids/syrups

- Corn syrup and high fructose corn syrup (HFCS)-40% fru, 50% glu
- Glucose syrup
- Honey
- Malt syrup
- Invert sugar
- Molasses

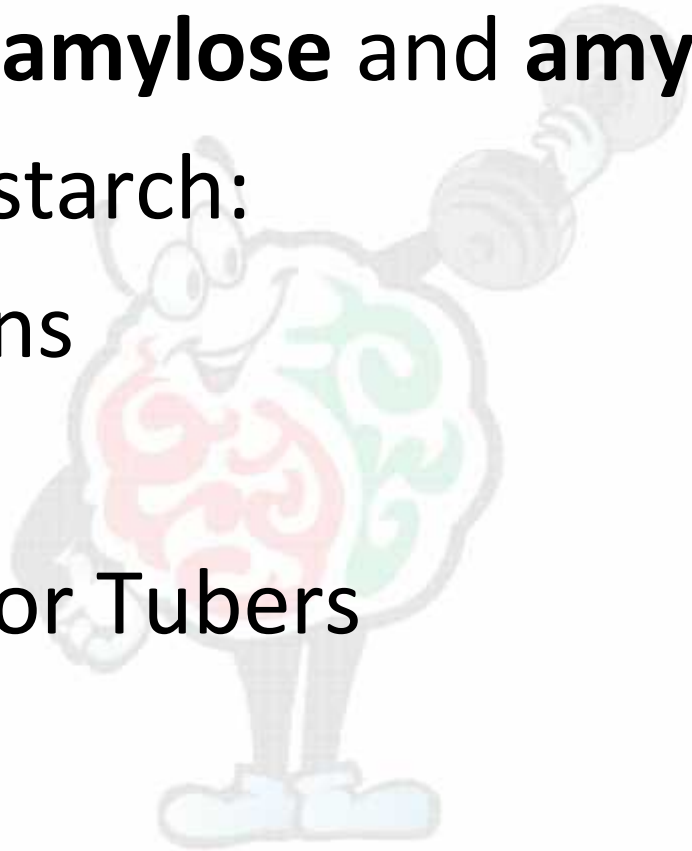


Characteristics of sugars:

1. Solubility: type & temperature
2. Ease of crystallization
3. Hygroscopic ability
4. Sweetness: type
5. Melting point
clear (160°C)---→caramel (170°C)
6. Hydrolysed by acid & enzyme
7. Fermentation: $\text{CO}_2 + \text{R-OH}$

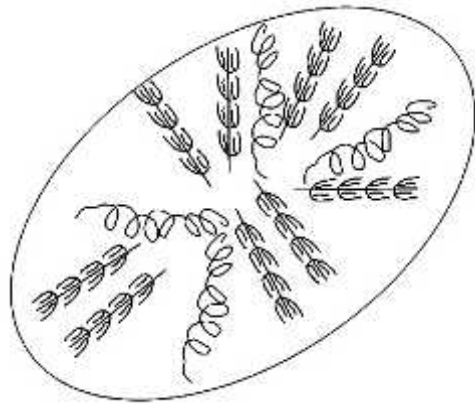
Starch

- Polysaccharide; made of two types of molecules: **amylose** and **amylopectin**
- Sources of starch:
 - Cereal grains
 - Legumes
 - Rootcrops or Tubers

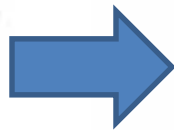


Effect of different types of treatment

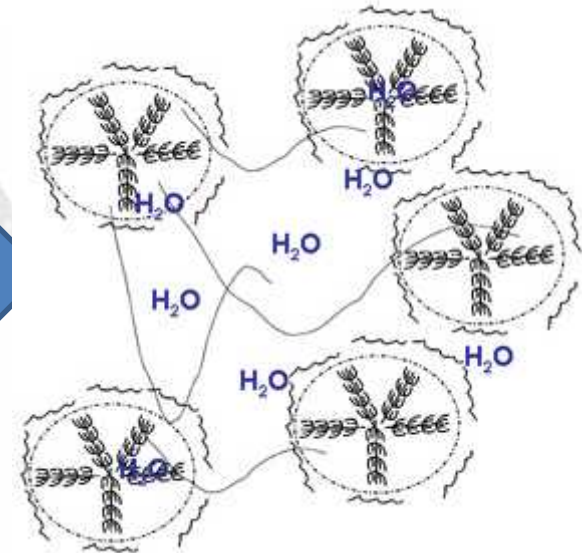
Treatment/ Ingredient	Changes & Effects
Dry heat	Dextrinization: non-enzymatic browning (texture, flavor, color and taste)
Moist heat	1. Gelatinization (88-92 °C) → sols (fluid paste) and gels (solid paste) 2. Gelation 3. Retrogradation 4. Syneresis



Starch in cold water



Water is absorbed and when it swells, amylose is released. Both molecules become less ordered



The starch granules form a matrix trapping water inside

Starch in Moist Heat



PECTIC SUBSTANCES

- Related carbohydrate derivatives that occur in plant tissues as **structural & cementing** substances
- Important in jelly-making
 1. Protopectin
 2. Pectinic acid (PECTIN)
 3. Pectic acid

(Brown, 2005 and de Leon, 1999 and Arroyo-Staub,1982)

CONVERSION OF PECTIC SUBSTANCES

PROTOPECTIN



**PECTINIC ACID
(PECTIN)**



PECTIC ACID

- Maturation
- Heat and acid

(Arroyo-Staub,1982)



Mango jelly



Mango jam



Mango marmalade



Mango preserves

